GROUNDLM WORKSHOP · EMNLP 2026

Compression-Aware Abstention

Teaching LLMs to refuse when KV-compression masks remove answer evidence

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Code, data & adapters
github.com/mali-kh/compression-aware-abstention

KV-cache compression can silently evict the evidence a question depends on. The model then **answers anyway — fluently and wrongly.** We train a **10M-parameter LoRA adapter** that answers when the evidence survives compression and refuses when it does not.

97%
fewer hallucinations
232 → 7 on KVzip-compressed MuSiQue

.32→.86
honest accuracy
base → ours, mean of 3 seeds

6–22×
more correct answers
under real compressed-cache decoding

96%
hallucination drop
replicates on Llama-3.1-8B (228 → 10)

1 Compression removes evidence. Models don't notice.

KV compressors (KVzip, SnapKV, Expected Attention) evict context tokens to save memory. When the evicted tokens carry the answer evidence, the remaining context is still **fluent and on-topic, but no longer determines the answer**, and the model fabricates one.

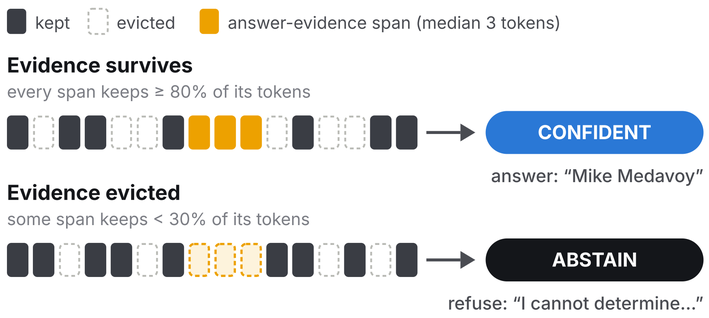
"Who founded the company that distributed the film *UHF*?"

Gold: Mike Medavoy · KVzip at r = 0.30 evicted the supporting tokens

✕ **BASE QWEN2.5-7B-INSTRUCT · HALLUCINATES**
"Kevin Smith founded the distribution company that released the film UHF."

✓ **+ OUR ADAPTER · ABSTAINS**
"I cannot determine the answer to this question from the available context."

2 Label by evidence survival, then train



1. **Tight-span localization.** MuSiQue 2-hop decompositions give each sub-answer as a short string; we locate it verbatim in its supporting paragraph (median **3 tokens**). Paragraph-level spans (~100 tokens) average out under compression and give zero label variance.
2. **Label from the mask.** Run a compressor at retention r ∈ {0.05 … 0.80}. **CONFIDENT** if every span keeps ≥ 80% of its tokens, **ABSTAIN** if any span keeps < 30%, otherwise dropped.
3. **Train.** Rank-16 LoRA (10.1M params, attention only) on frozen Qwen2.5-7B-Instruct: emit the gold answer on CONFIDENT, a fixed refusal on ABSTAIN.

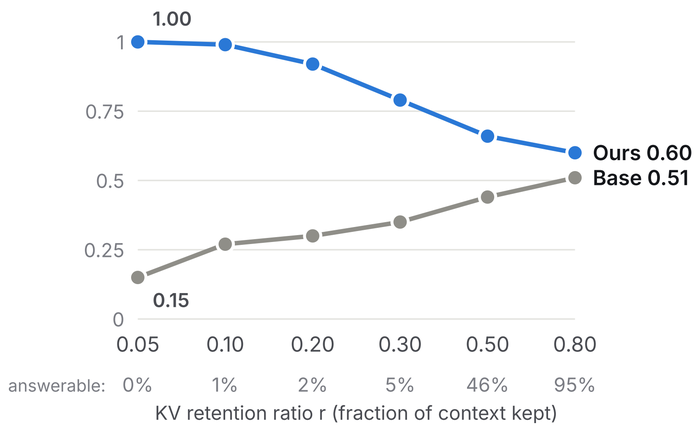
Honest accuracy = answer correctly when the evidence survived **+** refuse when it was evicted. Both directions count, so over-refusing is penalized as much as hallucinating.

Takeaway

Compressed models should know when they no longer know. A small adapter, trained on labels read off the compressor's own survival mask, turns silent evidence loss into an explicit refusal.

3 Hallucinations collapse

Honest accuracy on KVzip-compressed contexts, by retention ratio. The adapter refuses almost perfectly when evidence is gone and still answers when it survives.



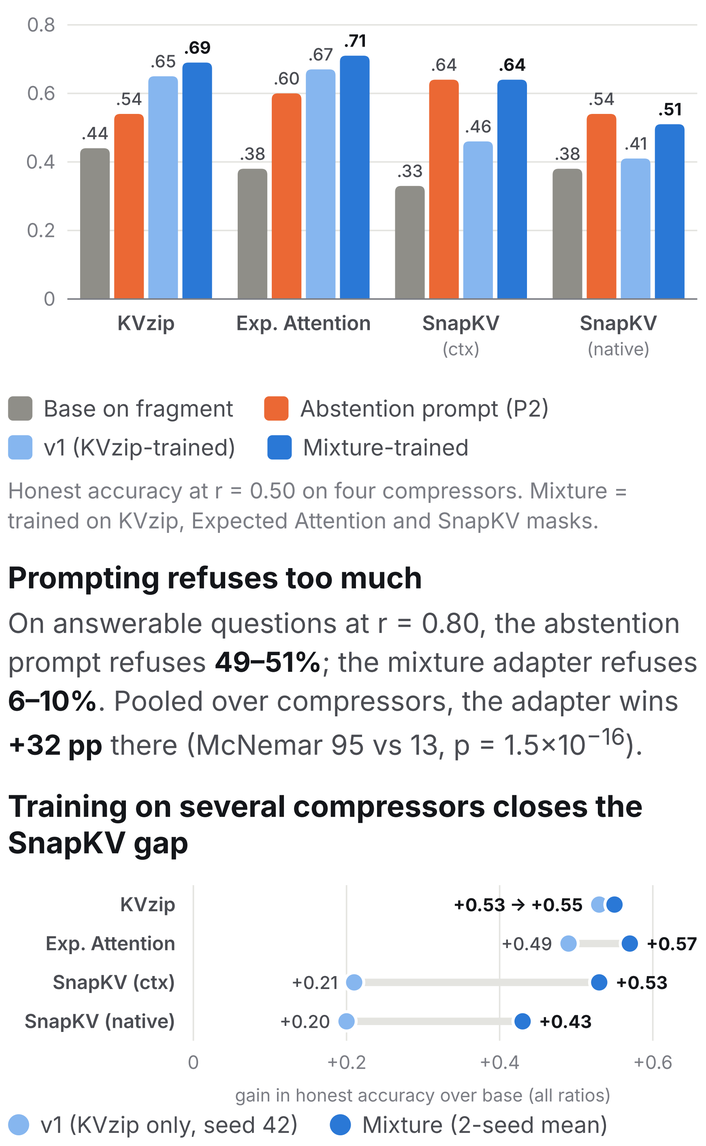
MuSiQue 2-hop validation, n = 463 (97 CONFIDENT / 366 ABSTAIN). "Answerable" = share of CONFIDENT examples at each ratio. Ours = mean of seeds 42/43/44.

232 → 7
hallucinated answers, base vs. adapter (seed 42)

1.00–1.36×
the base model's correct answers on answerable questions: safety without blanket refusal

4 Beats prompting; mixture training transfers across compressors

At **r = 0.50** about half the examples are answerable, so a rule based on r alone cannot help: the model has to use what survived.

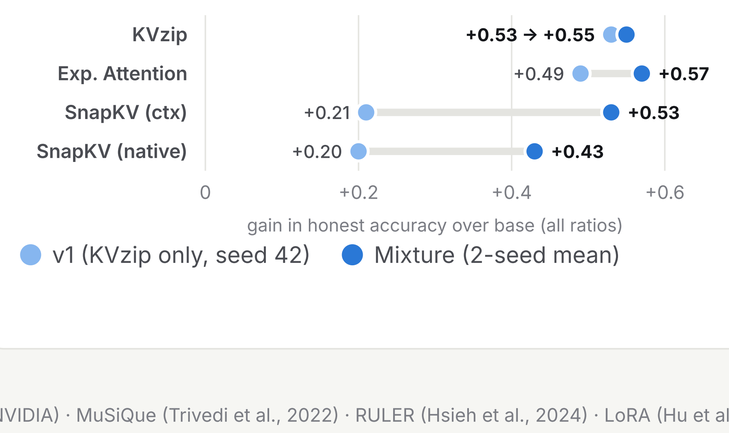


Honest accuracy at r = 0.50 on four compressors. Mixture = trained on KVzip, Expected Attention and SnapKV masks.

Prompting refuses too much

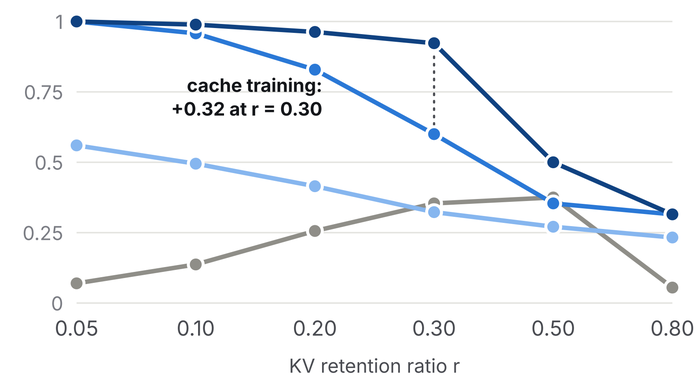
On answerable questions at r = 0.80, the abstention prompt refuses **49–51%**; the mixture adapter refuses **6–10%**. Pooled over compressors, the adapter wins **+32 pp** there (McNemar 95 vs 13, $p = 1.5\times10^{-16}$).

Training on several compressors closes the SnapKV gap



5 Works when decoding from the compressed cache

Sections 3–4 re-feed the kept tokens as text. Here the model decodes **directly from a KVzip-compressed cache**.

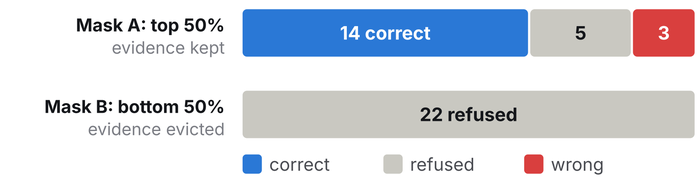


AGGREGATE HONEST ACCURACY, ALL RATIOS

Base model	**0.19**
v1 (prompt-style trained)	**0.41**
Mixture (prompt-style trained)	**0.73**
Mixture (cache-trained)	**0.82**

Correct answers on answerable questions: **24/97 vs. 4/97** for base (6×); **22/69 vs. 1/69** at r = 0.80 (22×).

6 Does it read the evidence?



Mixture adapter · 22 answerable questions at r = 0.50 · equal mask length

- **Delete only the evidence span** (same retention): refusals rise **16.5% → 56.7%**.
- Length is also a cue: fixing it at 500 tokens cuts CONFIDENT accuracy 0.62 → 0.15.

7 Generalizes beyond training

–38 pp
RULER needle retrieval at 16K (trained ≤ 4K, no retraining): hallucination 73% → 36%, correct answers unchanged

+0.64
Llama-3.1-8B replication: honest accuracy gain, hallucinations 228 → 10

Held-out test split: **+0.63** honest accuracy, hallucinations **59/85 → 1/85**.

8 Limitations

- **Aggregate scores can be gamed:** labels are U-shaped in r, so a rule on r alone reaches 0.93. We report per-ratio and per-class slices.
- **Span survival is a proxy** for sufficiency: κ = 0.61 vs. an LLM judge (80% agreement).
- Under compressed-cache decoding, answering at high retention is still hard (0.32 at r = 0.80).
- MuSiQue 2-hop only; compressed-cache tests use KVzipPress only.

References. KVzip (Kim et al., 2025) · SnapKV (Li et al., 2024) · Expected Attention / kvpress (NVIDIA) · MuSiQue (Trivedi et al., 2022) · RULER (Hsieh et al., 2024) · LoRA (Hu et al., 2022)

Adapters: huggingface.co/moalikhoda/compression-aware-abstention-adapters
Every number on this poster is recomputed by **scripts/verify_paper_claims.py**